

# Dynamics and Reconstruction in the Observable Measure

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## Abstract

The dynamics of observation, and underlying state space up to isometry, are derived from a  $\sigma$ -additive probability measure on the algebra of discriminable events [6]. The Rokhlin disintegration theorem (once  $\sigma$ -additivity holds) forces a conditional regularity kernel composed by the minimal sufficient factor  $Q_*$  carrying the coarsest full conditionally informative reduction. Indexing over time, temporal coherence forces the Chapman–Kolmogorov equation yielding a Markov semigroup  $\{\Pi_t\}$  and Koopman–Perron duality. In a measure-preserving system  $Q_*$  becomes the delay map  $\Phi_h$ , and reconstruction — that  $\mathcal{O}_h = \mathcal{B} \bmod \mu$ , density of the delay algebra in  $L^2$ , and injectivity of  $\Phi_h$  are equivalent — is the faithfulness question about this factor. The Stone space [6] built to derive  $\sigma$ -additivity is identified with  $X$  itself when reconstruction holds.

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## 1 Introduction

Koopman studied dynamical systems through observables [4], but the dynamical law  $T$  remains primitive: the operator  $U_T f = f \circ T$  is defined only after an underlying evolution has already been specified. The classical stochastic-process tradition proceeds similarly. Prediction is formulated relative to a filtration, so temporal order and information flow are built into the formalism from the outset [1]. That ordering is natural, but it need not be conceptually prior. One may instead ask whether observable probability already carries a conditional regularity between observables, and whether the familiar semigroup and predictive structures arise only when that regularity is organised coherently over time.

In [6],  $\sigma$ -additivity is not assumed but derived: a probability measure is forced into existence by the coherence of finite observations, at the cost of a single admissibility condition that no first-order structural requirement can supply. The measure is earned, not given. It does, however, already carry more than it was built to carry. Any two observations  $Q$  and  $F$  — no hypothesis on time or order — determine a conditional regularity of  $F$  given  $Q$ : the Markov kernel  $\kappa_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$ , which the Rokhlin disintegration theorem [7] forces into existence. It is not added; it is already there. Composing with  $Q$  gives the minimal sufficient factor  $Q_*$ , the coarsest reduction of the sample point that carries full conditional information about  $F$ . Temporal prediction is the special case in which  $F$  is a future observable; nothing in the construction requires it.

Index the conditional regularity over time and the kernels  $\{\Pi_t\}$  must compose consistently. That coherence forces Chapman–Kolmogorov — derived, not assumed — and from it a Markov semigroup and Koopman–Perron duality  $\int K_t g d\mu = \int g d(\mu \star \Pi_t)$ . In a measure-preserving system with  $Q_t = h \circ T^t$ , the factor  $Q_*$  becomes the delay map  $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$ , and the question becomes internal: does this factor see all of  $X$ , or only a proper quotient? The

answer — that  $\mathcal{O}_h = \mathcal{B} \bmod \mu$ , density of the delay algebra in  $L^2$ , and injectivity of  $\Phi_h$  are all equivalent — is the reconstruction theorem. The Stone space [6] built to derive  $\sigma$ -additivity is identified with  $X$  itself when reconstruction holds.

## 2 Conditional regularity and the canonical factor

### 2.1 The conditional regularity kernel

Let  $(\Omega, \mathcal{F}, P)$  be a  $\sigma$ -additive probability space on an observable algebra [6], and let  $Q, F$  be two measurable maps:

- $Q : \Omega \rightarrow \alpha$ , where  $(\alpha, \mathcal{A})$  is a measurable space;
- $F : \Omega \rightarrow \beta$ , where  $(\beta, \mathcal{B})$  is a standard Borel space.

The standard Borel hypothesis on  $\beta$  is used once: it is the hypothesis under which the Rokhlin disintegration theorem [7] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write  $\mathcal{P}(\beta)$  for the space of probability measures on  $\beta$ , equipped with its canonical (Giry)  $\sigma$ -algebra. The joint pushforward  $\rho := P \circ (Q, F)^{-1}$  on  $\alpha \times \beta$  has marginals  $\rho_\alpha = P \circ Q^{-1}$  and  $\rho_\beta = P \circ F^{-1}$ .

The regular conditional distribution of  $F$  given  $Q$  is the unique (up to  $P \circ Q^{-1}$ -null sets) Markov kernel  $\kappa_Q : \alpha \rightarrow \mathcal{P}(\beta)$  satisfying

$$\int_A \kappa_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable  $A \subseteq \alpha$  and  $B \subseteq \beta$ .

**Definition 2.1** (Conditional regularity kernel). The *conditional regularity kernel* of  $F$  given  $Q$  is  $\kappa_Q$ , the disintegration kernel of  $\rho = P \circ (Q, F)^{-1}$  over its first marginal:  $\kappa_Q(q, A) = P(F \in A \mid Q = q)$ . When  $F$  is a future observable the kernel describes conditional prediction; the object itself is neutral on time.

The kernel is covariant under measurable maps: if  $\pi : \alpha \rightarrow \alpha'$  is measurable and  $Q' = \pi \circ Q$ , then  $\kappa_Q(q, \cdot) = \kappa_{Q'}(\pi(q), \cdot)$  for  $(P \circ Q^{-1})$ -almost every  $q$ , by uniqueness of disintegration.

Composing  $\kappa_Q$  with  $Q$  gives the map sending each sample point to its conditional regularity.

### 2.2 The minimal sufficient factor

**Definition 2.2** (Minimal sufficient factor). The *minimal sufficient factor* is

$$Q_* : \Omega \rightarrow \mathcal{P}(\beta), \quad Q_*(\omega) = \kappa_Q(Q(\omega), \cdot).$$

Since  $\kappa_Q$  is a Markov kernel it is measurable, and  $Q_*$  inherits measurability by composition.

**Theorem 2.3** (Factorization via the canonical factor). *For every bounded measurable  $g : \beta \rightarrow \mathbb{R}$ :*

$$E[g(F) \mid \sigma(Q)] = \int g dQ_*(\cdot) \quad P\text{-a.e.}$$

*Proof.* Let  $h(\omega) := \int g dQ_*(\omega)$ . We verify the two defining properties.

*Measurability.* The map  $q \mapsto \int g d\kappa_Q(q, \cdot)$  is strongly measurable; since  $Q_* = \kappa_Q(Q(\cdot), \cdot)$ , the map  $h$  is  $\sigma(Q)$ -measurable.

*Set-integral identity.* For  $S = Q^{-1}(A)$ :

$$\begin{aligned} \int_S h dP &= \int_A \int g d\kappa_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{pushforward}) \\ &= \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration}) \\ &= \int_S g(F) dP. \end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation.  $\square$

Since  $\sigma(Q_*) \subseteq \sigma(Q)$  and the right-hand side is already  $\sigma(Q_*)$ -measurable,  $Q_*$  is a sufficient statistic for  $F$  given  $Q$ :  $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$   $P$ -a.e. It is *minimal* in the sense that every sufficient statistic factors through it:  $Q_*$  is the coarsest reduction of  $\omega$  that carries the full conditional regularity of  $F$  given  $Q$ . The operator  $(K_{Q_*}g)(q_*) = \int g dq_*$  on  $\mathcal{P}(\beta)$  recasts this as  $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$   $P$ -a.e.

### 3 Dynamics and duality

#### 3.1 Markov semigroup

When the conditional regularity kernel  $\kappa_Q$  is indexed over a time parameter, the kernels cannot be chosen independently: the regularity of  $F$  at time  $t + s$  given  $Q$  at time 0 must be consistent with first observing at time  $s$  and then at time  $t$ . This consistency requirement forces the Chapman–Kolmogorov equation.

Concretely, index over a discrete time horizon  $\mathbb{N}$  and let  $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$  be a Markov kernel on a state space  $\gamma$ , with  $\Pi_0(q, \cdot) = \delta_q$ . Temporal coherence of the conditional regularity demands

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s)$$

— the Chapman–Kolmogorov equation, derived from the structure rather than assumed. A family  $\{\Pi_t\}$  satisfying it is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Each  $K_t$  is positive, unital, and contractive on bounded measurable functions, with  $K_0 = \text{id}$ ; positivity, unitality, and contractiveness are immediate from the Markov property.

**Proposition 3.1** (Semigroup property).  $K_{t+s}g = K_t(K_s g)$  for all bounded measurable  $g$ .

*Proof.*  $(K_{t+s}g)(q) = \int g d(\Pi_t \circ_{\kappa} \Pi_s)(q, \cdot) = \int (\int g d\Pi_t(q', \cdot)) d\Pi_s(q, q') = \int (K_t g)(q') d\Pi_s(q, q') = (K_t(K_s g))(q)$ , where the exchange of integrals is justified by Bochner–Fubini.  $\square$

The pushforward of a measure  $\mu$  on  $\gamma$  through the kernel is  $P_t^* \mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot) d\mu(q)$ .

#### 3.2 Koopman–Perron duality

**Theorem 3.2** (Koopman–Perron duality). For every bounded measurable  $g$  and finite measure  $\mu$ :

$$\int (K_t g) d\mu = \int g d(P_t^* \mu).$$

*Proof.*  $\int g d(P_t^* \mu) = \int g d(\mu \star \Pi_t) = \int_{\gamma} \int g d\Pi_t(q, \cdot) d\mu(q) = \int (K_t g) d\mu$ , by Bochner–Fubini.  $\square$

When  $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$  for a measurable map  $\varphi_t : \gamma \rightarrow \gamma$ , the stochastic picture collapses.

### 3.3 Specialisation to measure-preserving systems

**Proposition 3.3** (Deterministic limit). *If  $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ , then  $K_t g = g \circ \varphi_t$  and  $\varphi_{t+s} = \varphi_t \circ \varphi_s$ .*

*Proof.*  $(K_t g)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$ . The flow law follows from  $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$  and injectivity of the Dirac embedding.  $\square$

In the deterministic case,  $K_t g = g \circ \varphi_t = U_T^t g$ : the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality  $\int K_t g d\mu = \int g d(P_t^* \mu)$  becomes the change-of-variables formula  $\int g \circ T^t d\mu = \int g d(T_*^t \mu)$ .

**Example 3.4** (Finite Markov chain). Let  $X = \{1, \dots, k\}$ ,  $\mathcal{B} = 2^X$ , and let  $P = (p_{ij})$  be the transition matrix of a finite irreducible Markov chain with unique stationary distribution  $\mu$  ( $\mu P = \mu$ ). Take full-state observation  $Q_t = X_t$ . The conditional regularity kernel of  $X_1$  given  $X_0$  is  $\kappa(i, \cdot) = P(i, \cdot)$ , the  $i$ -th row of  $P$ : this is the unique Markov kernel satisfying  $\mathbb{E}_\mu[f(X_1) \mid X_0 = i] = \sum_j p_{ij} f(j)$  for all bounded  $f$ . The  $t$ -step kernel  $\Pi_t(i, \cdot) = P^t(i, \cdot)$  is the corresponding semigroup kernel, and temporal coherence gives  $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$ , i.e.  $P^{t+s} = P^t P^s$ . The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_t g d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g d\mu,$$

the fixed-point property  $\mu P = \mu$  of the stationary distribution. In this finite setting every charge is automatically  $\sigma$ -additive, so the admissibility condition of [6] is trivially satisfied. Reconstruction holds at lag 0:  $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$  whenever  $h$  separates points of  $X$ .

The full structure produced by the construction is the triple  $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ : the canonical factor space, stationary distribution  $\nu_{Q_*} = P \circ Q_*^{-1}$ , and Markov semigroup. This is the *observable dynamical system* derived from  $(Q, F, P)$ : the conditional regularity of  $F$  relative to  $Q$ , promoted to a dynamical object by temporal coherence.

## 4 From regularity to reconstruction

Setting  $Q_t = h \circ T^t$  in a measure-preserving system  $(X, \mathcal{B}, \mu, T)$  with  $h \in L^\infty$  specialises the construction. Under this choice,  $Q_*(\omega) = (h(T^n \omega))_{n \in \mathbb{Z}}$  is the delay map  $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$ , and the observable dynamical system is the triple  $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$ .

The canonical factor space  $\Phi_h(X)$  is a subset of  $\mathbb{R}^{\mathbb{Z}}$  encoding the dynamical information that  $h$  can see. Whether it is a faithful image of  $X$  depends on how much the delay orbit  $\{h \circ T^n : n \in \mathbb{Z}\}$  collectively distinguishes points of  $X$ . Injectivity of  $\Phi_h$  modulo  $\mu$ -null sets is equivalent to the observable  $\sigma$ -algebra

$$\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$$

equalling  $\mathcal{B}(X)$  modulo  $\mu$ .

**Definition 4.1** (Faithful factor). A measurable map  $\psi : (X, \mathcal{B}, \mu) \rightarrow (Y, \mathcal{Y})$  is *faithful modulo  $\mu$*  if

$$\sigma(\psi) = \mathcal{B} \quad \text{modulo } \mu,$$

equivalently, if  $\psi$  separates  $\mu$ -a.e. pair of points of  $X$ . The reconstruction question is whether  $\Phi_h$  is faithful modulo  $\mu$ .

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable  $\sigma$ -algebra  $\mathcal{O}_h$  determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of  $\mathcal{B}$ .

#### 4.1 Separation defect and the role of CE

For a finite joined query algebra  $\mathcal{G} = \mathcal{E}_{n_1} \vee \cdots \vee \mathcal{E}_{n_k}$  from a directed system  $(\mathcal{E}_n)_n$ , the *separation defect* is

$$\delta(\mathcal{G}) := \sup_{A \in \sigma(\mathcal{C})} \inf_{E \in \mathcal{G}} \mu(A \triangle E),$$

where  $\sigma(\mathcal{C})$  is the observable  $\sigma$ -algebra of the full system. It vanishes iff  $\sigma(\mathcal{G}) = \sigma(\mathcal{C}) \bmod \mu$ . A sequence  $(\mathcal{G}_k)$  is *honest* if it is increasing and  $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$ ; under an honest scheme  $\delta(\mathcal{G}_k)$  is non-increasing.

**Theorem 4.2** (CE drives separation defect to zero). *Let  $(\mathcal{E}_n)_n$  be a query system with realization measure  $\mu$  satisfying collective exhaustion [6]. Under any honest refinement scheme  $(\mathcal{G}_k)$ ,*

$$\delta(\mathcal{G}_k) \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

*Proof.* Fix  $A \in \sigma(\mathcal{C})$ . Since  $(\mathcal{G}_k)$  is honest,  $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$ , so  $A$  is  $\sigma(\bigcup_k \mathcal{G}_k)$ -measurable. By the  $L^1(\mu)$  martingale convergence theorem,  $\mathbf{1}_{E_k} \rightarrow \mathbf{1}_A$  in  $L^1(\mu)$  where  $E_k = \mathbb{E}_\mu[\mathbf{1}_A \mid \mathcal{G}_k]$  rounded to the nearest atom; in particular  $\mu(A \triangle E_k) \rightarrow 0$ . CE [6] ensures no mass persists on the vanishing sets  $A \triangle E_k$ , confirming  $\inf_{E \in \mathcal{G}_k} \mu(A \triangle E) \rightarrow 0$  uniformly over  $A$ . Hence  $\delta(\mathcal{G}_k) \rightarrow 0$ .  $\square$

Sections 5–6 carry out the reconstruction in the delay-map setting.

**Remark 4.3** (The temporal specialisation). With  $Q_t = h \circ T^t$ , the algebra  $\mathcal{O}_h^{(L)}$  is an honest refinement scheme once reconstruction holds (§§5–6). The separation defect specialises to  $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ , the central quantity of [5]: exponential decay of  $\delta(L)$  under mixing yields minimax-optimal certification rates. In the general query setting, exhaustion and mixing are separate conditions; the delay-map setting collapses the first by construction.

## 5 Density bridge and reconstruction

### 5.1 Density bridge lemma

Throughout this section and the next,  $(X, \mathcal{B}, \mu)$  is a standard probability space,  $T : X \rightarrow X$  is a measurable,  $\mu$ -preserving invertible bimeasurable bijection with  $T_*\mu = \mu$ , and  $h \in L^\infty(X, \mu)$ .

**Definition 5.1** (Observable algebra). The *observable algebra* generated by  $h$  is

$$\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) \subseteq \mathcal{B}(X).$$

The *delay algebra*  $\mathcal{A}_h$  is the smallest subalgebra of  $L^\infty(X, \mu)$  containing 1 and every  $h \circ T^n$ ; concretely, finite sums  $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$ .

**Definition 5.2** (Delay map). The *delay map* is

$$\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}, \quad \Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}.$$

Since  $\pi_n \circ \Phi_h = h \circ T^n$  for each coordinate projection  $\pi_n$ , the map  $\Phi_h$  is measurable and  $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$ . Thus  $\mathcal{O}_h = \mathcal{B} \bmod \mu$  is precisely the condition that  $\Phi_h$  is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the  $L^2$  level. It holds for any subalgebra of  $L^\infty$ .

**Lemma 5.3** (Density bridge). *Let  $(X, \mathcal{B}, \mu)$  be a probability space and let  $\mathcal{A} \subseteq L^\infty(X, \mu)$  be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently,  $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$  if and only if  $\sigma(\mathcal{A}) = \mathcal{B}$  modulo  $\mu$ .

*Proof.* Write  $\mathcal{M} := \sigma(\mathcal{A})$  and  $V := \overline{\mathcal{A}}^{L^2}$ .

**Step 1:**  $V \subseteq L^2(X, \mathcal{M}, \mu)$ . Every  $f \in \mathcal{A}$  is  $\mathcal{A}$ -measurable, hence  $\mathcal{M}$ -measurable. So  $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$ . Since  $L^2(X, \mathcal{M}, \mu)$  is a closed subspace of  $L^2(X, \mathcal{B}, \mu)$ , taking the closure gives  $V \subseteq L^2(X, \mathcal{M}, \mu)$ .

**Step 2:**  $L^2(X, \mathcal{M}, \mu) \subseteq V$ . It suffices to show that every indicator  $\mathbf{1}_E$  with  $E \in \mathcal{M}$  lies in  $V$ . By the functional monotone class theorem [2], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing  $\mathcal{A}$  and closed under bounded monotone pointwise limits (since if  $f_n \in V$  are uniformly bounded and  $f_n \rightarrow f$  pointwise a.e. with  $f \in L^\infty$ , then  $f_n \rightarrow f$  in  $L^2$  by dominated convergence, so  $f \in V$ ). Since  $\mathcal{A}$  is an algebra containing 1 and  $\sigma(\mathcal{A}) = \mathcal{M}$ , the monotone class theorem gives  $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$ . In particular every  $\mathcal{M}$ -indicator  $\mathbf{1}_E \in V$ .

Since the  $\mathcal{M}$ -simple functions are dense in  $L^2(X, \mathcal{M}, \mu)$  and  $V$  is closed,  $L^2(X, \mathcal{M}, \mu) \subseteq V$ .

**Conclusion.** Steps 1 and 2 give  $V = L^2(X, \mathcal{M}, \mu)$ . The final equivalence is immediate:  $V = L^2(X, \mathcal{B}, \mu)$  iff  $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$  iff  $\mathcal{M} = \mathcal{B}$  mod  $\mu$ .  $\square$

**Remark 5.4** (Role of the algebra structure). The algebra structure of  $\mathcal{A}$  is essential. The corresponding statement for the closed linear span  $\mathcal{H}_{\mathcal{A}} = \overline{\text{span}}(\mathcal{A})$  is false in general:  $\mathcal{H}_{\mathcal{A}} = L^2$  does not imply  $\sigma(\mathcal{A}) = \mathcal{B}$ .

For example, take  $X = [0, 1]$ ,  $\mu = \text{Lebesgue}$ ,  $T(x) = 2x \bmod 1$  (the doubling map), and  $h(x) = x$ . The orbit  $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$  generates  $\mathcal{B}([0, 1])$  as a  $\sigma$ -algebra (the dyadic structure separates points). But  $\text{span}\{2^n x \bmod 1 : n \geq 0\}$  is a proper subspace of  $L^2[0, 1]$ : it does not contain  $x^2$ . So the linear span can be a proper dense subspace even when  $\sigma(\mathcal{A}) = \mathcal{B}$ .

In the other direction,  $h$  cyclic for  $U_T$  (linear span dense) always implies  $\sigma(\mathcal{A}_h) = \mathcal{B}$  mod  $\mu$  — see Corollary 6.2.

## 5.2 Reconstruction theorem

**Theorem 5.5** (Reconstruction theorem). *Let  $(X, \mathcal{B}, \mu, T)$  be an invertible measure-preserving system and  $h \in L^\infty(X, \mu)$ . The following are equivalent:*

- (i)  $\mathcal{O}_h = \mathcal{B}$  modulo  $\mu$ ;
- (ii)  $\mathcal{A}_h$  is dense in  $L^2(X, \mu)$ ;
- (iii)  $\Phi_h : X \rightarrow \mathbb{R}^\mathbb{Z}$  is a measure-theoretic embedding:  $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$  modulo  $\mu$ .

*Proof.* (i)  $\Leftrightarrow$  (ii). Apply the density bridge (Lemma 5.3) with  $\mathcal{A} = \mathcal{A}_h$  and  $\sigma(\mathcal{A}_h) = \mathcal{O}_h$ :  $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$  iff  $\mathcal{O}_h = \mathcal{B}$  mod  $\mu$ .

(i)  $\Leftrightarrow$  (iii). By Definition 5.2,  $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z}))$ . So  $\mathcal{O}_h = \mathcal{B}$  mod  $\mu$  iff  $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$  mod  $\mu$ , which is exactly condition (iii).  $\square$

Condition (iii) means  $\Phi_h$  induces an isomorphism of measure algebras  $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^\mathbb{Z}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^\mathbb{Z})/(\Phi_h)_*\mu)$ . Since  $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$ , the measure  $(\Phi_h)_*\mu$  is shift-invariant, and  $(X, T, \mu)$  is isomorphic to  $(\mathbb{R}^\mathbb{Z}, \sigma, (\Phi_h)_*\mu)$  restricted to  $\Phi_h(X)$ .

## 6 Cyclic vectors and Stone space identification

### 6.1 Cyclic vectors

**Definition 6.1** (Cyclic vector). The function  $h \in L^2(X, \mu)$  is a *cyclic vector* for  $U_T$  if

$$\mathcal{H}_h := \overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu).$$

**Corollary 6.2** (Cyclic implies reconstruction). *If  $h \in L^\infty(X, \mu)$  is a cyclic vector for  $U_T$ , then  $\mathcal{O}_h = \mathcal{B}$  modulo  $\mu$ .*

*Proof.* Since  $\mathcal{A}_h$  contains  $\text{span}\{h \circ T^n\} = \text{span}\{U_T^n h\}$  as a subset,  $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$ . By the density bridge,  $\mathcal{O}_h = \mathcal{B} \bmod \mu$ .  $\square$

**Remark 6.3** (The implication is strict). The converse of Corollary 6.2 fails in general:  $\mathcal{O}_h = \mathcal{B}$  does not imply  $h$  is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded  $h$ , the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when  $U_T$  has *simple  $L^2$ -spectrum*:  $L^2(X, \mu)$  is a cyclic  $U_T$ -module, meaning there exists some cyclic vector. In this case, every generating observation  $h$  (satisfying  $\mathcal{O}_h = \mathcal{B}$ ) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [3] but is not universal.

### 6.2 Stone space identification

The Stone space  $\text{St}(\mathcal{O}_h)$  of the observable algebra is the compact Hausdorff space whose points are ultrafilters on  $\mathcal{O}_h$ , with the natural Stone embedding  $\iota : X \rightarrow \text{St}(\mathcal{O}_h)$  sending  $x$  to the principal ultrafilter  $\{E \in \mathcal{O}_h : x \in E\}$ .

**Theorem 6.4** (Stone space identification). *Suppose  $\mathcal{O}_h = \mathcal{B}$  modulo  $\mu$ . Then  $\iota$  is a measure-theoretic isomorphism:  $\iota$  identifies the measure algebra  $(X/\mu, \mathcal{B}/\mu)$  with  $(\text{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\text{St}(\mathcal{O}_h))/\iota_*\mu)$ .*

*Proof.* *Injectivity mod  $\mu$ .* Suppose  $\iota(x) = \iota(x')$ . For any  $E \in \mathcal{B}$  with  $x \in E \not\equiv x'$ , find  $F \in \mathcal{O}_h$  with  $\mu(E \triangle F) = 0$ . Since  $x \in F$  and  $\iota(x) = \iota(x')$ , also  $x' \in F$ , so  $x' \in (E \triangle F) \cup F^c$ , placing  $x'$  in a null set. Hence  $\iota$  is injective  $\mu$ -a.e.

*$\sigma$ -algebra identification.* The pullback  $\iota^{-1}$  maps the Baire  $\sigma$ -algebra of  $\text{St}(\mathcal{O}_h)$  (generated by the clopen sets  $[E] = \{u : E \in u\}$ ) to  $\mathcal{O}_h = \mathcal{B} \bmod \mu$ .

*Measure recovery.* Since  $\mu$  is  $\sigma$ -additive on  $\mathcal{O}_h = \mathcal{B}$ , the collective exhaustion condition of [6] forces  $\iota_*\mu$  to concentrate on the principal ultrafilters  $\iota(X)$ , so  $\iota_*\mu$  is supported on the image and the measure algebras are identified.  $\square$

The Stone space [6] — built as a compact ambient space for deriving  $\sigma$ -additivity — is, under the reconstruction condition, the state space  $X$  itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

**Example 6.5** (Cantor set: reconstruction without smoothness). Let  $\mathcal{C} \subset [0, 1]$  be the middle-thirds Cantor set, and let  $\mu_{\mathcal{C}}$  be the Cantor measure — the pushforward of the fair-coin product measure  $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$  under the ternary coding map  $\pi(d_1, d_2, \dots) = \sum_{k \geq 1} d_k/3^k$ . Let  $S : \mathcal{C} \rightarrow \mathcal{C}$  be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let  $h(x) = x$ . The system  $(\mathcal{C}, \mu_{\mathcal{C}}, S)$  is measurably conjugate to the Bernoulli shift  $(\{0, 2\}^{\mathbb{N}}, \nu, \sigma)$  via  $\pi$ ; in particular  $S$  is  $\mu_{\mathcal{C}}$ -invariant and ergodic.

For  $x = \pi(d_1, d_2, \dots)$  we have  $S^n x = \pi(d_{n+1}, d_{n+2}, \dots)$ . Since every point of  $\mathcal{C}$  lies in  $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$ ,

$$d_{n+1} = 0 \iff S^n x \in [0, \frac{1}{3}], \quad d_{n+1} = 2 \iff S^n x \in [\frac{2}{3}, 1],$$

so  $h \circ S^n$  recovers the  $(n+1)$ -st ternary digit of  $x$ . The delay vector  $\Phi_h^{(L)}(x) = (x, Sx, \dots, S^L x)$  reads off digits  $d_1, \dots, d_{L+1}$ ; its fibres are the generation- $(L+1)$  cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k, 1 \leq k \leq L+1\}, \quad z \in \{0, 2\}^{L+1},$$

each carrying  $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$ , so  $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0, 2\}^{L+1}\}$ . Since  $\mu_{\mathcal{C}}$ -a.e. pair of distinct points differs at some finite digit, these algebras separate points as  $L \rightarrow \infty$ :

$$\mathcal{O}_h = \sigma\left(\bigcup_{L \geq 0} \mathcal{O}_h^{(L)}\right) = \mathcal{B}(\mathcal{C}) \mod \mu_{\mathcal{C}}.$$

By Theorem 5.5,  $\Phi_h$  is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as  $L \rightarrow \infty$ .

The observable algebra also encodes the geometry of  $\mathcal{C}$  directly. At each lag  $L$  there are  $2^{L+1}$  fibres of Euclidean diameter  $(1/3)^{L+1}$ , so the box-counting ratio  $\log 2^{L+1} / \log 3^{L+1} = \log 2 / \log 3$  is an algebraic invariant of the algebra itself, constant across all lags. The Hausdorff dimension is not a separate geometric calculation: it is read off from the refinement rate of the observable algebra.

$\mathcal{C}$  is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension  $\log 2 / \log 3 \approx 0.63$ . The differential-geometric hypotheses of Takens's theorem —  $C^2$  ambient dynamics, integer manifold dimension  $d$ , embedding bound  $2d+1$  — are unavailable before one even asks about regularity of  $S$ . Takens has no domain of application here. Theorem 5.5 applies because it requires only that the observable algebra be measurably generated; smoothness plays no role.

The classical Takens embedding theorem [8] reaches a related conclusion by a different route: for a  $C^2$  diffeomorphism on a compact  $d$ -manifold and a generic smooth  $h$ , the finite-delay map  $\Phi_h^{(k)}(x) = (h(x), h(Tx), \dots, h(T^{k-1}x))$  is a diffeomorphic embedding into  $\mathbb{R}^{2d+1}$  for  $k \geq 2d+1$ . The reconstruction theorem here requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth  $h$ , the orbit  $\{h \circ T^n\}$  separates points, so Stone–Weierstrass gives  $\mathcal{A}_h$  uniformly dense in  $C(M)$  and hence  $L^2$ -dense: the density condition holds and the two results are compatible. For non-smooth or zero-dimensional systems — the Cantor example being representative — the closer historical ancestor is the symbolic dynamics and generating-partition tradition [?]: representing orbits through successively finer finite partitions is exactly what the observable algebra  $\mathcal{O}_h^{(L)}$  does as  $L$  grows, with faithfulness of  $\Phi_h$  replacing the generating-partition condition.

**Remark 6.6** (Observable separation and Lyapunov exponents). When  $T$  is  $C^r$  and reconstruction holds, the delay map  $\Phi_h$  carries a natural instability rate: for  $\mu$ -a.e.  $x$  and nearby  $y$ ,

$$\lambda_h(x, y) := \limsup_{n \rightarrow \infty} \frac{1}{n} \log d(\Phi_h(T^n x), \Phi_h(T^n y)).$$

This *observable separation exponent* measures how fast the reconstructed dynamics distinguishes initial conditions through the observable, rather than how fast tangent vectors grow. When  $h$  is generic in the sense of Takens [8], the finite-lag embedding  $\Phi_h^{(L)}$  is bi-Lipschitz on the attractor,



sandwiching  $\lambda_h$  between the top Lyapunov exponent  $\lambda_1$  from above and below; one expects  $\lambda_h = \lambda_1$   $\mu$ -a.e., but making this precise requires comparing the induced metric on  $\Phi_h^{(L)}(X)$  with the ambient Riemannian geometry as  $L \rightarrow \infty$ , which is beyond the scope of the present paper. The quantitative geometry of fibre separation in this setting is taken up in [5].

## 7 Discussion

The measure derived in [6] already contains, without further assumption, the structures developed here. The Rokhlin disintegration reads off a conditional regularity kernel for any pair of observables; temporal indexing of that kernel forces Chapman–Kolmogorov and hence the Markov semigroup; and in a measure-preserving system the minimal sufficient factor becomes the delay map. Nothing is added to the measure — only consequences are drawn.

The reconstruction theorem closes a circle opened in [6]. The Stone space  $\text{St}(\mathcal{C})$  built there to establish  $\sigma$ -additivity is an abstract completion of the observable algebra, containing the realised state space as the principal ultrafilters but not yet identified with it. When  $\Phi_h$  is faithful modulo  $\mu$  (Definition 4.1), the three equivalent conditions of Theorem 5.5 jointly force  $\sigma(\Phi_h) = \mathcal{B}$  modulo  $\mu$ , and the Stone space collapses onto  $X$  itself: the abstract completion is the concrete state space. The question left open in [6] — whether prediction, dynamics, and state recovery follow from observation — is answered affirmatively, under faithfulness.

What faithfulness means in practice, and whether it can be certified from finite data, is taken up in [5].

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